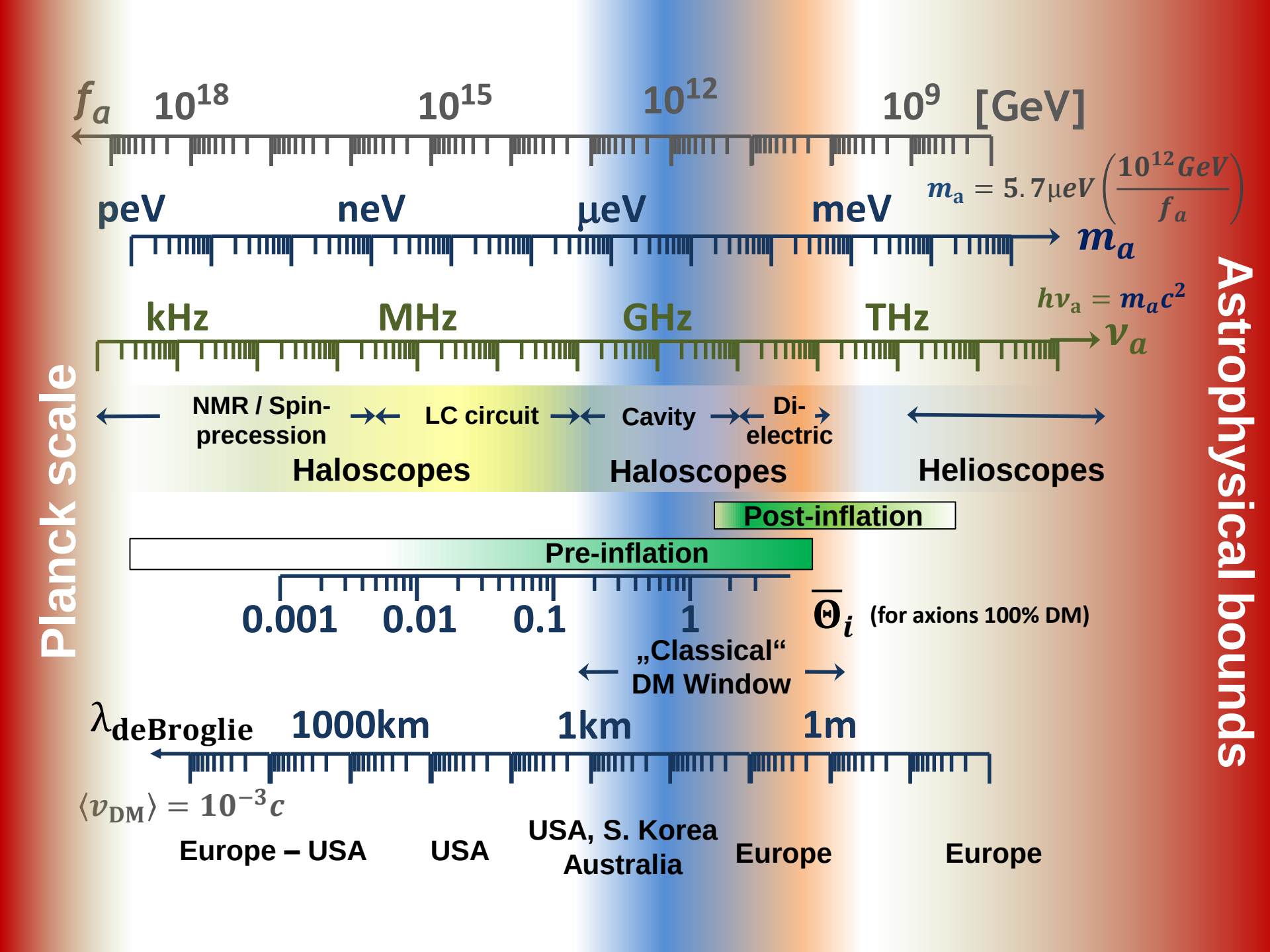
f_a 10^{18} 10^{15} 10^{12} 10^{9} [GeV]
$m_a = 5.7\mu eV\left(\frac{10^{12}GeV}{f_a}\right)$
peV neV μeV meV
m_a
$h\nu_a = m_a c^2$
kHz MHz GHz THz
ν_a
Planck scale
Astrophysical bounds
NMR / Spin-precession
LC circuit
Cavity
Di-electric
Haloscopes
Haloscopes
Helioscopes
Post-inflation
Pre-inflation
0.001 0.01 0.1 1
$\overline{\Theta}_i$ (for axions 100% DM)
„Classical“ DM Window
$\lambda_{deBroglie}$ 1000km 1km 1m
$\langle v_{DM} \rangle = 10^{-3}c$
Europe – USA
USA
USA, S. Korea Australia
Europe
Europe